Gig Cashflow Index (GCI) – 1 Month MVP

Implementation Plan

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1. Strategic & Regulatory Alignment Actions

- Define positioning as a Supplementary Cashflow Risk Index, not a credit bureau score.
- Align with RBI Digital Lending Guidelines and Account Aggregator consent framework.
- Prepare a policy note explaining how GCI complements bureau scores for gig workers.
- Design explainable AI outputs for lender transparency.
- Create a pilot proposal targeting NBFCs or MUDRA-aligned institutions.
- Ensure borrower consent, data encryption, and secure storage compliance.

2. Technical Stack & System Architecture

Core Stack

- **Frontend:** React (lightweight dashboard for borrower & lender view).
- **Backend:** FastAPI (Python) for API and model serving.
- **Database:** PostgreSQL (structured financial records).
- **ML Framework:** Scikit-learn (Logistic Regression / Gradient Boosting).
- **Data Processing:** Pandas & NumPy.
- **Authentication:** JWT-based secure login.
- **Deployment:** Docker + AWS EC2 / Render.
- **Security:** Data Storage Encryption and HTTPS communication mandatory.

Basic System Architecture Flow

Borrower Upload / AA API → Data Cleaning Layer → Transaction Classification Engine → Feature Engineering Module → **[Manipulation Detection Module]** → Risk Model (PD Prediction) → Stress Simulation Engine → Gig Cashflow Index Generator → Explainability Layer → Dashboard Output

Note: Manipulation Detection Module is newly inserted between Feature Engineering and Risk Model to intercept and penalise anomalous transaction profiles before scoring.

3. One-Month Development Roadmap

Week 1: Foundation & Data Layer

- Finalize feature definitions (income volatility, liquidity ratio, EMI ratio).
- Create synthetic dataset (6–12 months transaction simulation).
- Build transaction classification logic.
- Design PostgreSQL schema for storing structured cashflow data.

Week 2: Modeling & Simulation

• Implement feature engineering pipeline.

• Train initial Probability of Default model.

• Develop stress simulation module (20% & 30% income drop scenarios).

• Validate outputs on synthetic cases.

■ *Manipulation Detection Module (New — Week 2 Addition)*

Implement an Isolation Forest-based anomaly detection module on engineered cashflow features. Train on synthetic clean profiles and inject 5 synthetic manipulation patterns to validate detection sensitivity. Define contamination threshold and map anomaly scores to GCI penalty multipliers.

Manipulation Patterns the Module Defends Against

• **Income Spiking:** Artificially inflated inflows in 60–90 days before loan application (friends/family round-tripping).

• **Expense Suppression:** Sudden disappearance of recurring expenses (rent, subscriptions, recharges) in the assessment window.

• **Round-Trip Transactions:** Same amount credited and debited to/from the same UPI counterparty within 48–72 hours.

• **Synthetic Regularity:** Highly irregular income profile suddenly showing perfectly consistent weekly inflows — statistically too clean.

• **Dormant Account Activation:** Account with minimal historical activity suddenly showing heavy transaction volume in scoring window.

Algorithm Stack

• **Primary — Isolation Forest (Unsupervised)**

– Works by randomly partitioning the feature space; anomalous points get isolated in very few splits.

– Ideal for MVP: unsupervised, no labeled fraud data required, efficient on tabular cashflow features.

– Contamination parameter set at 5–10%; tunable as real data accumulates.

– Anomaly scores mapped to a GCI penalty multiplier (Suspicious: 0.75x | High Risk: 0.50x).

• **Secondary — Statistical Rules Layer (Hard Rules, Non-ML)**

– Flag if income in last 90 days exceeds 2 standard deviations above borrower's own 12-month mean.

– Flag round-trip transactions: same amount in/out from same UPI ID within 72 hours.

– Flag if recurring expense signatures visible in historical data disappear in the assessment window.

– Flag inbound transfers from non-merchant, non-employer sources exceeding threshold in assessment window.

• **Post-MVP Upgrade — Autoencoder (Shallow Neural Network)**

– Trained on 'normal' gig transaction patterns to learn a compressed latent representation.

– High reconstruction error on manipulated profiles becomes the anomaly score.

– Catches subtle non-linear manipulation patterns that Isolation Forest may miss.

Features Fed Into Isolation Forest

• Month-over-month income change rate (spike detection).

• Ratio of inflow to outflow per week over rolling windows.

• Std deviation of weekly inflows: 6-month historical vs last 30 days (sudden regularity flag).

• Count of round-trip transaction pairs within 72-hour windows.

• Dormancy score: transaction frequency delta between historical and assessment window.

• Expense-to-income ratio trend over rolling 3-month periods.

Output of Manipulation Detection Module

Manipulation Risk Label: Clean | Suspicious | High Risk + **Penalty Multiplier:** 1.0x | 0.75x | 0.50x applied to GCI score before output. Lender dashboard displays the label, anomaly score, and primary flag reason for full explainability and RBI compliance.

Week 3: Backend & Interface

- Build FastAPI backend endpoints.
- Develop Gig Cashflow Index calculation logic.
- Integrate explainability (feature contribution display).
- Design simple React dashboard.
- **Expose Manipulation Risk Label and anomaly score via dedicated API endpoint.** Display manipulation flag with brief explainability (e.g., *'Unusual income spike detected in assessment window'*) in the lender dashboard view.

Week 4: Deployment & Validation

- Deploy MVP on cloud.
- Run 10–20 simulated borrower case studies.
- Prepare policy brief + demo presentation for lenders or MUDRA alignment.
- **Manipulation Detection Testing (Expanded):** Test across 5 synthetic attack scenarios:
  - Income spiking (3x average inflow in last 60 days).
  - Round-trip transactions (same amount in/out within 48 hours).
  - Expense suppression (known recurring charges disappear).
  - Synthetic regularity (irregular historical profile suddenly becomes perfectly weekly).
  - Dormant account activation (near-zero activity then sudden heavy usage).
- Measure and document false positive rate on clean profiles. Tune contamination threshold accordingly.

4. Expected MVP Output

- Gig Cashflow Index score (300–900 scale).
- Probability of Default estimate.
- Income Stability Score.
- Liquidity Buffer Score.
- Stress Survival Analysis Output.
- Explainability Report for lender review.
- **Manipulation Risk Label + Anomaly Score + Primary Flag Reason** (lender-facing, explainable, RBI-compliant).

Manipulation Detection Module added per GCI design review. Isolation Forest (primary) + Statistical Rules Layer (hard flags) + Autoencoder (post-MVP). All flags are graduated and transparent — no black-box rejection.